

SMOOTH EXPANDING MAPS

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1. THE PROBLEM

Consider the map $T \in \mathcal{C}^2(\mathbb{T}^2, \mathbb{T}^2)$ such that $|DT| \geq \lambda > 1$. We would like to know if there exists measures absolutely continuous with respect to Lebesgue and, if so, how many. The idea is to study the transfer operator

$$(1.1) \quad \mathcal{L}h(x) := \sum_{y \in T^{-1}x} |D_y T|^{-1} h(y)$$

Exercise 1. Show that if $d\mu = hdm$, where m is the Lebesgue measure, then $\mu(f \circ T) = m(f\mathcal{L}h)$.

Exercise 2. Show that, for each $n \in \mathbb{N}$,

$$\mathcal{L}^n h(x) := \sum_{y \in T^{-n}x} |D_y T^n|^{-1} h(y)$$

Notice that, since DT cannot be zero, then its sign is constant. We can then assume without loss of generality, $DT \geq \lambda$.

Exercise 3. Show that

$$\begin{aligned} \frac{d}{dx} \mathcal{L}^n h(x) &= \sum_{y \in T^{-n}x} (D_y T)^{-2} h'(y) - D_y^2 T (D_y T)^{-3} h(y) \\ &= \mathcal{L}((DT)^{-1} h') - \mathcal{L}(D^2 T (DT)^{-2} h) \end{aligned}$$

2. A FUNCTIONAL ANALYTIC SETTING

Let us consider first the Sobolev space $W^{1,1}$ and the space L^1 .¹ Then, for each $h \in L^1(\mathbb{T}^1, m)$,

$$(2.1) \quad \int_{\mathbb{T}^1} |\mathcal{L}h| dm \leq \int_{\mathbb{T}^1} 1 \cdot \mathcal{L}|h| dm = \int_{\mathbb{T}^1} 1 \circ T |h| dm = \int_{\mathbb{T}^1} |h| dm$$

that is $\mathcal{L}$ is a bounded operator on L^1 and its norm is bounded by one.

In addition, remembering Exercise 1,

$$(2.2) \quad \int_{\mathbb{T}^1} \left| \frac{d}{dx} \mathcal{L}h \right| dm \leq \lambda^{-1} |h'|_{L^1} + D |h|_{L^1},$$

where $D := \sup D^2 T (DT)^{-2}$.

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¹For an open set $U \subset \mathbb{R}$, the spaces $W^{p,q}(U)$ are the completion of $\mathcal{C}^\infty(U, \mathbb{C})$ with respect to the norms $[|f|_{L^q}^q + |f'|_{L^q}^q + \dots + |f^{(p)}|_{L^q}^q]^{\frac{1}{q}}$. Note that they are all Banach spaces by construction but the $W^{p,2}$ are also Hilbert spaces (**Exercise:** write the scalar product).

Exercise 4. Iterate the (2.1), (2.2) and prove, for all $n \in \mathbb{N}$,

$$\begin{aligned} |\mathcal{L}^n h|_{L^1} &\leq |h|_{L^1} \\ |\mathcal{L}^n h|_{W^{1,1}} &\leq \lambda^{-n} |h|_{W^{1,1}} + B |h|_{L^1} \end{aligned}$$

where $B = 1 + (1 - \lambda^{-1})^{-1} D$.

Since $W_{1,1}$ controls the L^∞ norm,² then we have that there exists $C > 0$ such that $|\mathcal{L}^n 1|_\infty < C$ for each $n \in \mathbb{N}$.

Using such a fact we can obtain similar inequalities in the Hilbert spaces L^2 and $W^{1,2}$. Indeed

$$\begin{aligned} \|\mathcal{L}^n h\|_{L^2}^2 &= \int_{\mathbb{T}^1} h(\mathcal{L}^n h) \circ T^n \leq \|h\|_{L^2} \left[\int_{\mathbb{T}^1} (\mathcal{L}^n h)^2 \circ T^n \right]^{\frac{1}{2}} = \|h\|_{L^2} \left[\int_{\mathbb{T}^1} (\mathcal{L}^n h)^2 \mathcal{L}^n 1 \right]^{\frac{1}{2}} \\ &\leq C^{\frac{1}{2}} \|h\|_{L^2} \|\mathcal{L}^n h\|_{L^2} \end{aligned}$$

Which implies $\|\mathcal{L}^n h\|_{L^2} \leq C^{\frac{1}{2}} \|h\|_{L^2}$ for each $n \in \mathbb{N}$. Hence,

$$\left\| \frac{d}{dx} \mathcal{L}^n h \right\|_{L^2} \leq \lambda^{-n} C^{\frac{1}{2}} \|h'\|_{L^2} + D_n \|h\|_{L^2}.$$

Iterating as before we have, for all $n \in \mathbb{N}$,

$$\begin{aligned} |\mathcal{L}^n h|_{L^2} &\leq C |h|_{L^2} \\ |\mathcal{L}^n h|_{W^{1,2}} &\leq A \lambda^{-n} |h|_{W^{1,2}} + B |h|_{L^2}, \end{aligned} \tag{2.3}$$

for some appropriate constants A, B, C depending only on the map T .

To prove the existence of an invariant measure absolutely continuous with respect to Lebesgue we can try to mimic the Krylov-Bogolubov approach, but to do so we need a compactness result to substitute the weak compactness of the unit ball of the dual of a Banach space. This takes us in a very interesting detour in some fact of functional analysis.

3. DEEPER IN FUNCTIONAL ANALYSIS

Since we are on a circle it is a good idea to use Fourier series. For each function $h \in \mathcal{C}^\infty(\mathbb{T}, \mathbb{C})$ let h_k be its Fourier coefficients and define

$$(3.1) \quad (\mathbb{A}_m h)(x) = \sum_{|k| \leq m} h_k e^{2\pi i k x}$$

Clearly, for all $m > 0$,

$$\begin{aligned} |h - \mathbb{A}_m|_{L^2}^2 &= \sum_{|k| > m} |h_k|^2 = \sum_{|k| > m} |h_k|^2 |k|^{-2} |k|^2 \leq m^{-2} \sum_{|k| > m} |(h')_k|^2 \\ &\leq m^{-2} |h'|_{L^2}^2 \leq m^{-2} |h|_{W^{1,2}}^2. \end{aligned} \tag{3.2}$$

Using the above fact we can prove.

Lemma 3.1. *The unit ball of $W^{1,2}$ is (sequentially) compact in L^2 .*

²If $f \in \mathcal{C}^\infty$, then the mean value theorem asserts $\int h = h(\xi)$ for some ξ . Then $h(x) = h(\xi) + \int_\xi^x h'(z) dz$. Thus $|h|_\infty \leq |h|_{L^1} + |h'|_{L^1} = |h|_{W^{1,1}}$. The result extends then to all elements of $W^{1,1}$ by a standard approximation argument.

Proof. Consider a sequence $\{h_m\} \subset W^{1,2}$, $|h_m|_{W^{1,2}} \leq 1$. Since $\mathbb{A}_l$ are all finite rank operators, $\{\mathbb{A}_l h_m\}$ for l fixed are contained in a bounded finite dimensional (hence compact) set, thus there exists a converging subsequence for all l while (3.2) shows that the sequences for fixed m are all convergent. Using the usual diagonalization trick we can then extract a converging subsequence. $\square$

Consider now $h_n := \frac{1}{n} \sum_{k=0}^{n-1} \mathcal{L}^k 1$. By the above lemma $\{h_n\}$ is relatively compact and thus we can extract a subsequence $\{h_{n_j}\}$ converging in L^2 . Let h_* be the limit. Note that $\int h_n = 1$ for all $n \in \mathbb{N}$, thus $h_* \neq 0$ and $\int h_* = 1$.

Exercise 5. Show that $\mathcal{L}h_* = h_*$, that is $d\mu := h_* dm$ is an invariant measure absolutely continuous with respect to Lebesgue and with L^2 density.

Of course, at this point it is natural to ask if μ is the only measure with such a property or there exist others. To answer such a question we need some more facts.

4. EVEN DEEPER IN FUNCTIONAL ANALYSIS

Since we have to do it, let us do in the following general setting.

Consider two Banach space $(\mathcal{B}, \|\cdot\|)$ and $(\mathcal{B}_0, |\cdot|)$ such that $\mathcal{B} \subset \mathcal{B}_0$ and

- i. $|h| \leq \|h\|$ for all $h \in \mathcal{B}$,
- ii. if $h \in \mathcal{B}$ and $|h| = 0$, then $h = 0$.
- iii. There exists $C > 0$: for each $\varepsilon > 0$ there exists a finite rank operator $\mathbb{A}_\varepsilon \in L(\mathcal{B}, \mathcal{B})$ such that $\|\mathbb{A}_\varepsilon\| \leq C$ and $|h - \mathbb{A}_\varepsilon h| \leq \varepsilon \|h\|$ for all $h \in \mathcal{B}$.³

In addition consider a bounded operator $\mathcal{L} : \mathcal{B}_0 \rightarrow \mathcal{B}_0$, constants $A, B, C \in \mathbb{R}_+$, and $\lambda > 1$, such that

- a. $|\mathcal{L}^n| \leq C$ for all $n \in \mathbb{N}$,
- b. $\mathcal{L}(\mathcal{B}) \subset \mathcal{B}$
- c. $\|\mathcal{L}^n h\| \leq A\lambda^{-n}\|h\| + B|h|$ for all $h \in \mathcal{B}$ and $n \in \mathbb{N}$.

In particular $\mathcal{L}$ can be seen as a bounded operator on $\mathcal{B}$.

Theorem 4.1. *The spectral radius of the operator $\mathcal{L} \in L(\mathcal{B}, \mathcal{B})$ is bounded by 1 while the essential spectral radius is bounded by λ^{-1} .⁴*

The proof of the above theorem depends on an auxiliary fact which is of much interest in its own.

Theorem 4.2 (Analytic Fredholm theorem–finite rank⁵). *Let D be an open connected subset of $\mathbb{C}$. Let $F : \mathbb{C} \rightarrow L(\mathcal{B}, \mathcal{B})$ be an analytic operator-valued function such that $F(z)$ is finite rank for each $z \in D$. Then, one of the following two alternatives holds true*

³In fact, this last property can be weakened to: The unit ball $\{h \in \mathcal{B} : \|h\| \leq 1\}$ is relatively compact in $\mathcal{B}_0$. We use the present stronger condition since, on the one hand, it is true in all the applications we will be interested in and, on the other hand, drastically simplifies the argument. Note also that, if one uses the Fredholm alternative for compact operators rather than finite rank ones (Theorem 4.2), then one can ask the $\mathbb{A}_\varepsilon$ to be compact instead than finite rank making easier their construction in concrete cases.

⁴The definition of *essential spectrum* varies a bit from book to book. Here we call essential spectrum the complement, in the spectrum, of the isolated eigenvalues with associated finite dimensional eigenspaces (which is also called the Fredholm spectrum).

⁵The present proof is patterned after the proof of the Analytic Fredholm alternative for compact operators (in Hilbert spaces) given in [1, Theorem VI.14]. There it is used the fact that compact operators in Hilbert spaces can always be approximated by finite rank ones. In fact the theorem holds also for compact operators in Banach spaces but the proof is a bit more involved.

- $(\mathbf{Id} - F(z))^{-1}$ exists for no $z \in D$
- $(\mathbf{Id} - F(z))^{-1}$ exists for all $z \in D \setminus S$ where S is a discrete subset of D (i.e. S has no limit points in D). In addition, if $z \in S$, then 1 is an eigenvalue for $F(z)$ and the associated eigenspace has finite multiplicity.

Proof. First of all notice that, for each $z_0 \in D$ there exists $r > 0$ such that $D_{r(z_0)}(z_0) := \{z \in \mathbb{C} : |z - z_0| < r(z_0)\} \subset D$, and

$$\sup_{z \in D_{r(z_0)}(z_0)} \|F(z) - F(z_0)\| \leq \frac{1}{2}.$$

Clearly if we can prove the theorem in each such disk we are done.⁶ Note that

$$\mathbf{Id} - F(z) = (\mathbf{Id} - F(z_0)(\mathbf{Id} - [F(z) - F(z_0)])^{-1})(\mathbf{Id} - [F(z) - F(z_0)]).$$

Thus the invertibility of $\mathbf{Id} - F(z)$ in $D_r(z_0)$ depends on the invertibility of $\mathbf{Id} - F(z_0)(\mathbf{Id} - [F(z) - F(z_0)])^{-1}$. Let us set $F_0(z) := F(z_0)(\mathbf{Id} - [F(z) - F(z_0)])^{-1}$.

Let us start by looking at the equation

$$(4.1) \quad (\mathbf{Id} - F_0(z))h = 0.$$

Clearly if a solution exists, then $h \in \text{Range}(F_0(z)) = \text{Range}(F(z_0)) := \mathbb{V}_0$. Since $\mathbb{V}_0$ is finite dimensional there exists a basis $\{h_i\}_{i=1}^N$ such that $h = \sum_i \alpha_i h_i$. On the other hand there exists an analytic matrix $G(z)$ such that⁷

$$F_0(z)h = \sum_{ij} G(z)_{ij} \alpha_j h_i.$$

Thus (4.1) is equivalent to

$$(\mathbf{Id} - G(z))\alpha = 0,$$

where $\alpha := (\alpha_i)$.

The above equation can be satisfied only if $\det(\mathbf{Id} - G(z)) = 0$ but the determinant is analytic hence it is either always zero or zero only at isolated points.⁸

Suppose the determinant different from zero, and consider the equation

$$(\mathbf{Id} - F_0(z))h = g.$$

Let us look for a solution of the type $h = \sum_i \alpha_i h_i + g$. Substituting yields

$$\alpha - G(z)\alpha = \beta$$

where $\beta := (\beta_i)$ with $F_0(z)g =: \sum_i \beta_i h_i$. Since the above equation admits a solution, we have $\text{Range}(\mathbf{Id} - F_0(z)) = \mathcal{B}$. Thus we have an everywhere defined inverse, hence bounded by the open mapping theorem.

⁶In fact, consider any connected compact set K contained in D . Let us suppose that for each $z_0 \in K$ we have a disk $D_{r(z_0)}(z_0)$ in the theorem holds. Since the disks $D_{r(z_0)/2}(z_0)$ form a covering for K we can extract a finite cover. If the first alternative holds in one such disk then, by connectness, it must hold on all K . Otherwise each $S \cap D_{r(z_0)/2}(z_0)$, and hence $K \cap S$, contains only finitely many points. The Theorem follows by the arbitrariness of K .

⁷To see the analyticity notice that we can construct linear functionals $\{\ell_i\}$ on $\mathbb{V}_0$ such that $\ell_i(h_j) = \delta_{ij}$ and then extend them to all $\mathcal{B}$ by the Hahn-Banach theorem. Accordingly, $G(z)_{ij} := \ell_j(F_0(z)h_i)$, which is obviously analytic.

⁸The attentive reader has certainly noticed that this is the turning point of the theorem: the discreteness of S is reduced to the discreteness of the zeroes of an appropriate analytic function: a determinant. A moment thought will immediately explain the effort made by many mathematicians to extend the notion of determinant (that is to define an analytic function whose zeroes coincide with the spectrum of the operator) beyond the realm of matrices (the so called Fredholm determinants).

We are thus left with the analysis of the situation $z \in S$ in the second alternative. In such a case, there exists h such that $(\mathbf{Id} - F(z))h = 0$, thus one is an eigenvalue. On the other hand, if we apply the above facts to the function $\Phi(\zeta) := \zeta^{-1}F(z)$ analytic in the domain $\{\zeta \neq 0\}$ we note that the first alternative cannot take place since for $|\zeta|$ large enough $\mathbf{Id} - \Phi(\zeta)$ is obviously invertible. Hence, the spectrum of $F(z)$ is discrete and can accumulate only at zero. This means that there is a small neighborhood around one in which $F(z)$ has no other eigenvalues, we can thus surround one with a small circle γ and consider the projector

$$\begin{aligned} P &:= \frac{1}{2\pi i} \int_{\gamma} (\zeta - F(z))^{-1} d\zeta = \frac{1}{2\pi i} \int_{\gamma} [(\zeta - F(z))^{-1} - \zeta^{-1}] d\zeta \\ &= \frac{1}{2\pi i} \int_{\gamma} F(z) \zeta^{-1} (\zeta - F(z))^{-1} d\zeta. \end{aligned}$$

By standard functional calculus it follows that P is a projector and it clearly projects on the eigenspace of the eigenvector one. But the last formula shows that P must project on a subspace of the range of $F(z)$, hence it must be finite dimensional. $\square$

We can now prove our main result.

Proof of Theorem 4.1. The first assertion is a trivial consequence of (c), (a) and (i).

The second part is much deeper. Let $\mathcal{L}_{n,\varepsilon} := \mathcal{L}^n \mathbb{A}_{\varepsilon}$, clearly such an operator is finite rank, in addition

$$\|\mathcal{L}^n h - \mathcal{L}_{n,\varepsilon} h\| \leq A\lambda^{-n} \|(\mathbf{Id} - \mathbb{A}_{\varepsilon})h\| + B\|(\mathbf{Id} - \mathbb{A}_{\varepsilon})h\| \leq A(1+C)\lambda^{-n} \|h\| + B\varepsilon \|h\|.$$

By choosing $\varepsilon = \lambda^{-n}$ we have that there exists $C_1 > 0$ such that

$$\|\mathcal{L}^n - \mathcal{L}_{n,\varepsilon}\| \leq C_1 \lambda^{-n}.$$

For each $z \in \mathbb{C}$ we can now write

$$\mathbf{Id} - z\mathcal{L} = (\mathbf{Id} - z(\mathcal{L} - \mathcal{L}_{n,\varepsilon})) - z\mathcal{L}_{n,\varepsilon}.$$

Since

$$\|z(\mathcal{L} - \mathcal{L}_{n,\varepsilon})\| \leq |z|C_1\lambda^{-n} < \frac{1}{2},$$

provided that $|z| \leq \frac{1}{2C_1}\lambda^n$. Thus, given any z in the disk $D_n := \{|z| < \frac{1}{2C_1}\lambda^n\}$ the operator $B(z) := \mathbf{Id} - z(\mathcal{L} - \mathcal{L}_{n,\varepsilon})$ is invertible.⁹ Hence

$$\mathbf{Id} - z\mathcal{L} = (\mathbf{Id} - z\mathcal{L}_{n,\varepsilon}B(z)^{-1})B(z) = (\mathbf{Id} - F(z))B(z).$$

By applying Theorem 4.2 to $F(z)$ we have that the operator is either never invertible or not invertible only in finitely many points in the disk D_n . Since for $|z| < 1$ we have $(\mathbf{Id} - z\mathcal{L})^{-1} = \sum_{n=0}^{\infty} z^n \mathcal{L}^n$, the first alternative cannot hold hence the Theorem follows. $\square$

⁹Clearly $B(z)^{-1} = \sum_{n=0}^{\infty} [z(\mathcal{L} - \mathcal{L}_{n,\varepsilon})]^n$.

5. THE HARVEST

We are finally in the position to use all the above result to gain a deep understanding of the properties of the Dynamical Systems under consideration.

Exercise 6. Show that Theorem 4.1 implies that there exists $\sigma \in (0, 1)$, $\{\theta_k\}_{k=1}^p$ and $L > 0$ such that

$$\mathcal{L} = \sum_{k=1}^p e^{i\theta_k} \Pi_{\theta_k} + R$$

where Π_{θ_k} and R are operators on $W^{1,2}$ such that $\Pi_{\theta_k} \Pi_{\theta_j} = \delta_{jk} \Pi_{\theta_k}$ and $R \Pi_{\theta_k} = \Pi_{\theta_k} R = 0$. Moreover $|R^n| \leq L \sigma^n$. (Hint: Read section 6 of the Third Chapter of [2] and recall that the operator is power bounded to exclude Jordan blocks.)

The above implies that

$$(5.1) \quad \Pi_\theta := \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} e^{-i\theta k} \mathcal{L}^k = \begin{cases} \Pi_{\theta_i} & \text{iff } \theta = \theta_j \\ 0 & \text{otherwise.} \end{cases}$$

Exercise 7. Using equations (2.3) show that, for each $h \in L^2$

$$\|\Pi_\theta h\|_{W^{1,2}} \leq C \|h\|_{L^2}.$$

(Hint: prove it first for $h \in W^{1,2}$ and then do a density argument).

Next, note that Exercise 5 implies that $h_* = \Pi_0 1 \neq 0$, that is one is in the spectrum on $\mathcal{L}$, this means that the spectral radius of $\mathcal{L}$ is one.

Accordingly, if $\Pi_\theta h = h$ we have $h \in W^{1,2} \subset \mathcal{C}^0$ and¹⁰

$$|h| = |\Pi_\theta h| \leq \lim_{j \rightarrow \infty} \frac{1}{n_j} \sum_{k=0}^{n_j-1} \mathcal{L}^k |h| = \Pi_0 |h| \leq |h|_\infty h_*.$$

This means that all the eigenvectors of the peripheral spectrum are of the form $h = g h_*$ with $g \in \mathcal{C}^0$. Thus, if h_i is an $W^{1,2}$ orthonormal a base of the eigenspace associated to an eigenvalue θ , then the eigenprojector must have the form

$$\Pi_\theta h = \sum_i h_i \int \ell_i \cdot h,$$

with $\ell_i \in L^2$ and $\int \ell_i h_j = \delta_{ij}$. Hence $\Pi_\theta \mathcal{L} = e^{i\theta} \Pi_\theta$ implies

$$e^{i\theta} \sum_k h_k \int \ell_k \cdot h = \sum_k h_k \int \ell_k \cdot \mathcal{L} h = \sum_k h_k \int \ell_k \circ T \cdot h.$$

That is $e^{i\theta} \ell_k = \ell_k \circ T$. But then if we set $f_k := \bar{\ell}_k h_* \in L^2$, we have

$$\mathcal{L} f_k = e^{i\theta} \mathcal{L}(\bar{\ell}_k \circ T h_*) = e^{i\theta} \bar{\ell}_k \mathcal{L} h_* = e^{i\theta} \bar{\ell}_k h_* = e^{i\theta} f_k$$

By the above facts, this implies $\Pi_\theta f_k = f_k \in W^{1,2}$, that is $\ell_k \in \mathcal{C}^0$. But this implies that if we define the sets $A_L := \{x \in \mathbb{T} : |\ell_k| \leq L\}$, $L \in \mathbb{R}$, they are all invariant. So if χ_L is the characteristic function of the set A_L , then $\chi_L \circ T = \chi_L$ and $\mathcal{L}(\chi_L h_*) = \chi_L h_*$. We can thus produce a lot of eigenvalues of $\mathcal{L}$, but we know that such eigenvalues form a finite dimensional space. The only possibility is that only finitely many of the A_L are different. This is like saying that ℓ_k takes only

¹⁰Remember that exercise 7 implies that the sequence in (5.1) converges in L^2 , accordingly there exists a subsequence that converges almost everywhere with respect to Lebesgue.

finitely many values. But ℓ_k is a continuous function, so it must be constant. In turn this implies $\theta = 0$.

The conclusion is that one is the only eigenvalue on the unit circle and that the associated eigenprojector has rank one. So one is a simple eigenvalue and h_* is the only invariant density for the map.

6. CONCLUSIONS

If we have any probability measure ν absolutely continuous with respect to Lebesgue and with density $h \in W^{1,2}$, then setting $d\mu = h_* dm$, for each $\varphi \in W^{1,2}$ we have

$$|\mu(\varphi \circ T^n) - \nu(\varphi \circ T^n)| = \left| \int \varphi \mathcal{L}^n(h - h_*) \right| \leq \|\varphi\|_{1,2} C \sigma^n \|h - h_*\|_{1,2}$$

where σ is the largest eigenvalue of modulus smaller than one (or λ^{-1} is no such eigenvalue exist).

The above means that the evolution of a chaotic system, if seen at the level of the absolutely continuous measures, becomes simply a dynamics with an attracting fixed point, the simplest possibility of all!

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